04/11/26

Signal Processing

- Recap: given $|H(e^{j\omega})|$, how does $\angle H(e^{j\omega})$ look?
- $C(z) = H(z) \cdot H^*(1/z^*)$; $|H(e^{j\omega})|^2 = C(z)\Big|_{z=e^{j\omega}}$
- Relating magnitude and phase response
  - | |, x, o; ∠, x, o, min-phase
  - $|H(e^{j\omega})|^2$
  - Examples

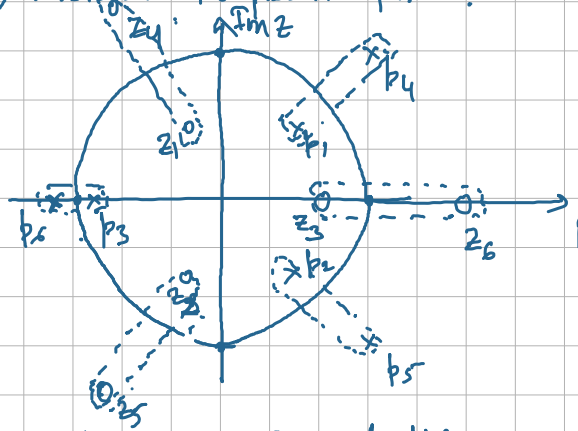
1: • $H_1(z) = \left[2(1-z^{-1})(1+0.5z^{-1})\right] / \left[(1-0.8e^{j\pi/4}z^{-1})(1-0.8e^{j\pi/4}z^{-1})\right]$

• $H_2(z) = \left[(1-z^{-1})(1+2z^{-1})\right] / \left[(1-0.8e^{j\pi/4}z^{-1})(1-0.8e^{-j\pi/4}z^{-1})\right]$

• $C_1(z) = H_1(z) \cdot H_1^*(1/z^*) = \left(\frac{2(1-z^{-1})(1+0.5z^{-1})}{(1-0.8e^{j\pi/4}z^{-1})(1-0.8e^{-j\pi/4}z^{-1})}\right)\left(\frac{2(1-z^{-1})(1+2z^{-1})}{(1-\frac{e^{-j\pi/4}}{0.8}z^{-1})(1+\frac{e^{j\pi/4}}{0.8}z^{-1})}\right)^*$

2: $C(z)$ with poles and zeros marked below. If system is stable, causal, and a_k, b_k are real, what are the possible options?

- stable → poles must be inside the unit circle
- causal → poles: p_1, p_2, p_3
- a_k, b_k are real → pole, zero at conjugate locations



z_3 or z_6

(z_1, z_2) or (z_4, z_5)

- Recall: $H(z) = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^{M}(1-c_k z^{-1})}{\prod_{k=1}^{N}(1-d_k z^{-1})}$ ⟸ (rational system)
  - Assuming b_0, a_0 are real.
  - $\prod_{k=1}^{M}(1-c_k z^{-1})$: zero of $H(z)$
  - $\prod_{k=1}^{N}(1-d_k z^{-1})$: poles of $H(z)$
- Ideal delay: $H_{id}(e^{j\omega}) = e^{-j\omega\alpha}$, $|\omega| < \pi$ ⟸ α is real
- Ideal LPF with linear-phase: $H_{lp}(e^{j\omega}) = \begin{cases} e^{-j\omega\alpha}, & |\omega| < \omega_c \\ 0, & \omega_c < |\omega| \le \pi \end{cases}$, symmetry

- Generalized linear-phase system: $H(e^{j\omega}) = A(e^{j\omega}) \cdot e^{-j(\omega\alpha-\beta)}$; $H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n}$

- Let's now look at LTI systems (filters) with linear phase response

  - $H_{id}(e^{j\omega}) = e^{-j\omega\alpha}$, $0 < |\omega| < \pi$

    $|H_{id}(e^{j\omega})| = 1$ —— 1 ——

    $\angle H_{id}(e^{j\omega}) = -\omega\alpha$ ——— ——

    $grad[H_{id}(e^{j\omega})] = \alpha$ ——— ———

$$h_{id}[n] = \frac{\sin \pi(n-\alpha)}{\pi(n-\alpha)} \qquad -\infty < n < \infty \qquad \text{(IIR filters)}$$

If $\alpha = n_d$ (integer)

$h_{id}[n] = \delta[n-n_d]$ } $y[n] = x[n] * h_{id}[n]$

- $H_{lp}(e^{j\omega}) = \begin{cases} e^{-j\omega\alpha}, & |\omega| < \omega_c \\ 0, & \omega_c < |\omega| \le \pi \end{cases}$ Low pass with linear phase response

$$h_{lp}[n] = \frac{\sin \omega_c(n-\alpha)}{\pi(n-\alpha)} \quad (?) \qquad \text{(IIR)}$$

- $H_{lp}(e^{j\omega}) = |H_{lp}(e^{j\omega})| \cdot \exp\left(\angle H_{lp}(e^{j\omega})\right)$ (another representation)

- Let's say $\alpha = n_d$. Can we say something about $h_{lp}[n]$?

$h_{lp}[2n_d - n] = h_{lp}[n]$ (Symmetry of $h_{lp}[n]$)